RAJALAKSHMI ENGINEERING COLLEGE

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N2

Started on	Saturday, 5 September 2026, 9:28 AM
State	Finished
Completed on	Saturday, 5 September 2026, 9:29 AM
Time taken	35 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Write a Program to Implement the Quick Sort Algorithm

Input Format:
The first line contains the no of elements in the list-n
The next n lines contain the elements.

Output:
Sorted list of elements

For example:

Input	Result
5 67 34 12 98 78	12 34 67 78 98

Answer:

```c
#include <stdio.h>

void quickSort(int a[], int low, int high)
{
    if (low < high)
    {
        int i = low, j = high;
        int pivot = a[(low + high) / 2];

        while (i <= j)
        {
            while (a[i] < pivot)
                i++;

            while (a[j] > pivot)
                j--;

            if (i <= j)
            {
                int temp = a[i];
                a[i] = a[j];
                a[j] = temp;
                i++;
                j--;
            }
        }

        if (low < j)
            quickSort(a, low, j);

        if (i < high)
            quickSort(a, i, high);
    }
}

int main()
{
    int n;
    scanf("%d", &n);

    int a[n];

    for (int i = 0; i < n; i++)
        scanf("%d", &a[i]);

    quickSort(a, 0, n - 1);

    for (int i = 0; i < n; i++)
    {
        if (i > 0)
            printf(" ");
```

```
52 |         printf("%d", a[i]);
```

	Input	Expected	Got	
✔	5 67 34 12 98 78	12 34 67 78 98	12 34 67 78 98	✔
✔	10 1 56 78 90 32 56 11 10 90 114	1 10 11 32 56 56 78 90 90 114	1 10 11 32 56 56 78 90 90 114	✔
✔	12 9 8 7 6 5 4 3 2 1 10 11 90	1 2 3 4 5 6 7 8 9 10 11 90	1 2 3 4 5 6 7 8 9 10 11 90	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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